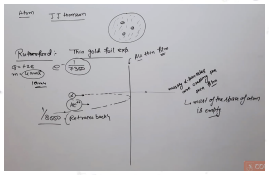
Atoms and Bohr's Model

→ J.J Thompson suggested the plum pudding model. According to him atom is a sphere with all the volume filled with +ve charge and -ve charge is embedded in it in form of $e^{\ominus}$.



Rutherford {1906 - 1911} : (exp)

→ He conducted an experiment called Thin gold foil experiment.
→ He took a very thin foil of called and shooted α-particles on it.
→ Most of the α-particles crossed the foil but 1/8000 retraced back.

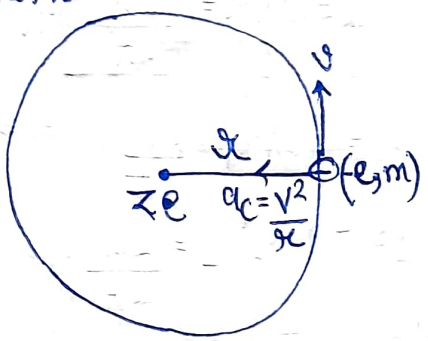
Rutherford's postulates:

→ Most of the space of atom is empty.
→ All the neutrons and protons are concentrated in very small sphere known as nucleus.
→ Nucleus is center of atom.
→ $e^{\ominus}$ will revolve around the nucleus and columbic force b/w $e^{\ominus}$ and nucleus will provide the required centripetal acceleration.
→ Nucleus has practically all the mass of atom. It consists of neutrons and protons, together called nucleons and has entire +ve charge of atom.
Radius of Nucleus ≈ 1 Fermi = 10^{-15} m
→ Separation b/w nucleus and $e^{\ominus}$ is radius of atom, $\mathring{A} = 10^{-10}$ m

$$\frac{\text{Nucleus}}{\text{atom}} = 10^{-5}$$

Calculations:

Z = no. of protons



$$\frac{Ze^2}{4\pi\varepsilon_0 r^2} = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{Ze^2}{4\pi\varepsilon_0 m r}}$$

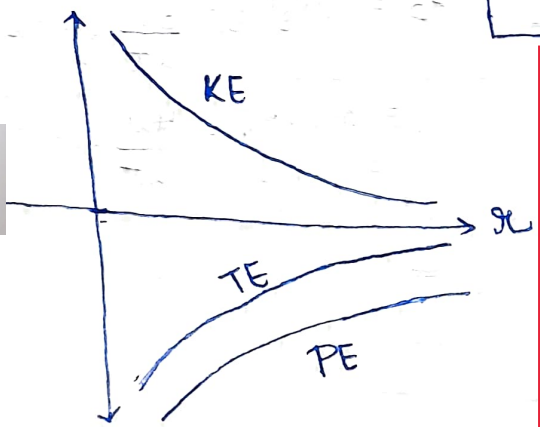
$$KE = \frac{1}{2}mv^2 = \frac{Ze^2}{8\pi\varepsilon_0 r}$$

$$PE = \frac{-Ze^2}{4\pi\varepsilon_0 r}$$

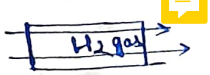
$$T.E. = \frac{-Ze^2}{8\pi\varepsilon_0 r}$$

{-ve sign indicates that system is bounded}

Graph:



Q. Why Rutherford failed?
① Maxwell suggested that charge particle with variable velocity will emit EM wave due to which energy ↓ and $e^{\ominus}$ will collapse in nucleus.
② In H-spectrum experiment, only a specific value of λ was absorbed but acc. to rutherford $e^{\ominus}$ should have emitted all λ



$$\lambda = \frac{364.56m^2}{m^2-4}\ nm$$

m = 3, 4, 5 . . . . .

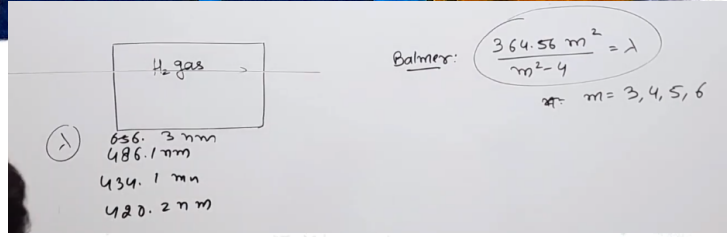
Bohr's Model:

Postulates:-

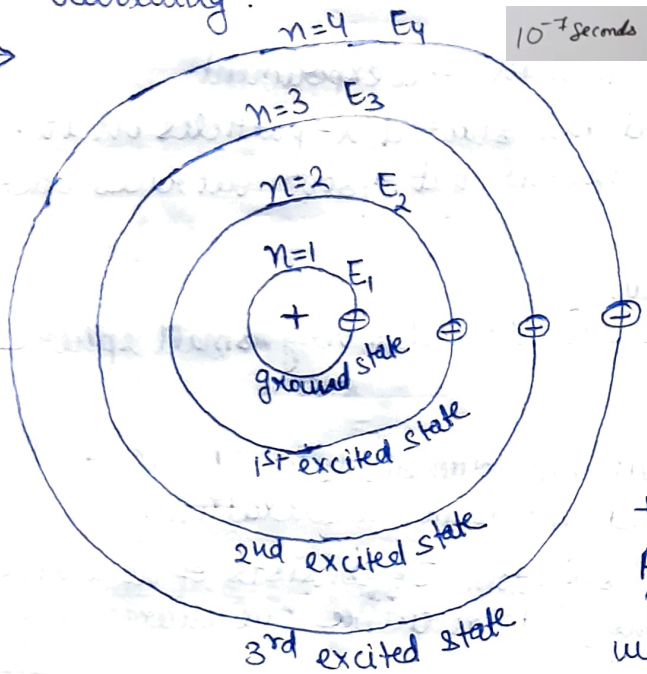
→ Nucleus is center of atom

→ $e^{\ominus}$ will revolve around atom

→ $e^{\ominus}$ will revolve in specific orbits, known as stationary orbits. And $e^{\ominus}$ revolving in these stationary orbits doesn't emit any radiation while revolving.



→



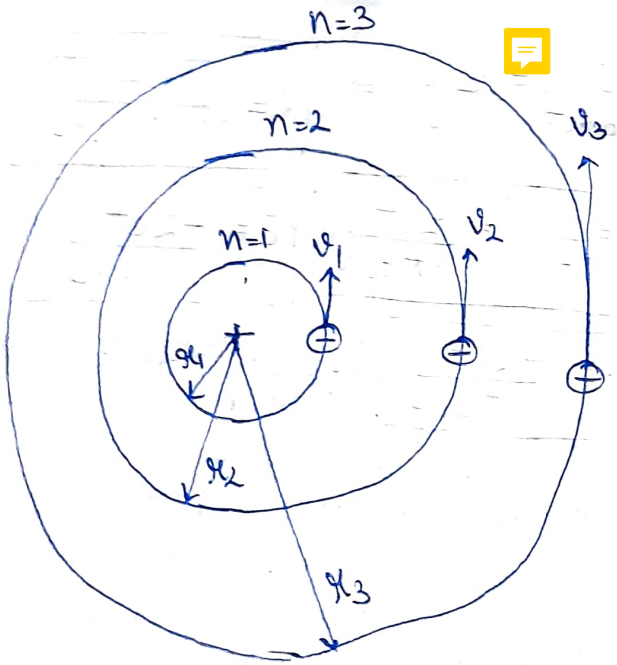
10^{-7} seconds

An $e^{\ominus}$ in any stationary orbit can only absorb a specific energy / λ of photon.

If a photon with energy as $E_2 - E_1 = \frac{hc}{\lambda}$ strikes the $e^{\ominus}$ in ground state, then this $e^{\ominus}$ will absorb this photon and make a transition to n=2. Suppose energy is $E_4 - E_1 = \frac{hc}{\lambda}$, then $e^{\ominus}$ will jump to 3rd excited state ie n=4.

Also when this $e^{\ominus}$ comes back from n=4 to n=1, it will radiate photon with energy $E_4 - E_1 = \frac{hc}{\lambda}$.

→ Bohr's Quantisation rule: It states that angular momentum of $e^{\ominus}$ about nucleus is quantized and always integral multiple of $h/2\pi$.



$$L = mvr = \frac{nh}{2\pi} \quad n = 1, 2, 3, 4 \ldots$$

$L_1 = mv_1r_1 = \frac{h}{2\pi}$ {n=1 ⇒ ground state}

$L_2 = mv_2r_2 = \frac{2h}{2\pi}$ {n=2 ⇒ 1st excited state}

$L_3 = mv_3r_3 = \frac{3h}{2\pi}$ {n=3 ⇒ 2nd excited state}

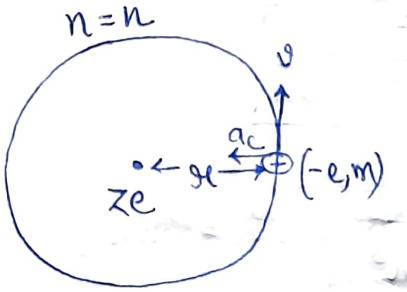
$L_4 = mv_4r_4 = \frac{4h}{2\pi}$ {n=4 ⇒ 3rd excited state}

.
.
.

Calculations by Bohr's Postulates:

For H-like atom (only $1e^{\ominus}$) ⇒ H, He^{+}, Li^{+2}, Be^{+3}, B^{+4} etc.

No. of protons = 1, 2, 3, 4, 5
(Z)



NL: $\frac{Ze^2}{4\pi\varepsilon_0 r^2} = \frac{mv^2}{r} \Rightarrow \frac{Ze^2}{4\pi\varepsilon_0 r} = mv^2$ — (1)

AMQR: $L = mvr = \frac{nh}{2\pi}$ — (2)

n = Stationary orbit no. / Quantum no.

n = 1, 2, 3, 4 . . . .

K L M N . . . .

① Radius of orbit:

$$\frac{Ze^2}{4\pi\varepsilon_0 r} = m\,\frac{n^2h^2}{4\pi^2 m^2 r^2} \Rightarrow$$

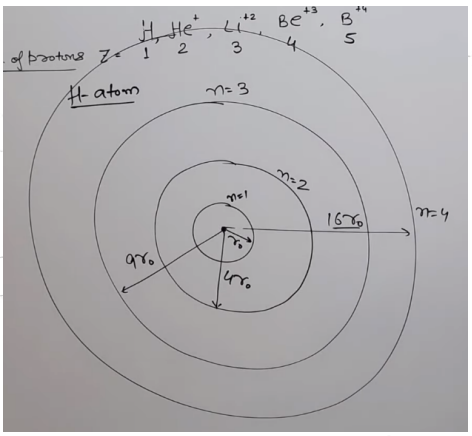
$$\boxed{r = \frac{h^2\varepsilon_0}{\pi me^2}\left(\frac{n^2}{Z}\right)}$$

$$\boxed{r = 0.529\,\text{Å}\left(\frac{n^2}{Z}\right) = r_0\frac{n^2}{Z}}$$

r_0 = Bohr's radius

$$\boxed{\begin{aligned} r &\propto n^2 \\ r &\propto \frac{1}{Z} \end{aligned}}$$

	H	He^{+}	Li^{+2}	Be^{+3}
Z	1	2	3	4
n=1	r_0	$r_0/2$	$r_0/3$	$r_0/4$
n=2	$4r_0$	$2r_0$	$4r_0/3$	r_0
n=3	$9r_0$	$9r_0/2$	$3r_0$	$9r_0/4$
n=4	$16r_0$	$8r_0$	$16r_0/3$	$4r_0$



② Speed of $e^{\ominus}$:

$$v = \frac{nh}{2\pi mr} = \frac{nh\pi me^2}{2\pi n h^2 \varepsilon_0}\left(\frac{Z}{n^2}\right)$$

$$\boxed{v = \left(\frac{e^2}{2h\varepsilon_0}\right)\frac{Z}{n}}$$

$$\boxed{v = 2.19\times10^6\ \text{m/s}\left(\frac{Z}{n}\right)}$$

$$\boxed{\begin{aligned} v &\propto Z \\ v &\propto \frac{1}{n} \end{aligned}}$$

③ Angular speed of $e^{\ominus}$:

$$\omega = \frac{v}{r} = \frac{e^2}{2h\varepsilon_0}\frac{m e^2}{h^2\varepsilon_0}\left(\frac{Z}{n}\right)\left(\frac{Z}{n^2}\right)$$

$$\omega = \frac{\pi m e^4}{2h^3\varepsilon_0^2}\left(\frac{Z^2}{n^3}\right)$$

$$\omega = 4.14\times10^{16}\,\frac{Z^2}{n^3}\ \text{rad/s}$$

$$\star\ \omega \propto Z^2,\quad \omega \propto \frac{1}{n^3}$$

④ Time Period of $e^{\ominus}$:

$$T = \frac{2\pi}{\omega} = \frac{2\pi\, 2h^3\varepsilon_0^2}{\pi m e^4}\left(\frac{n^3}{Z^2}\right)$$

$$T = \frac{4h^3\varepsilon_0^2}{me^4}\left(\frac{n^3}{Z^2}\right)$$

$$T = 1.5\times10^{-16}\left(\frac{n^3}{Z^2}\right)\text{ sec}$$

$$\star\ T \propto n^3,\quad T \propto \frac{1}{Z^2}$$

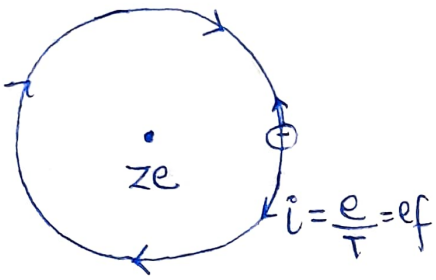
⑤ Frequency of $e^{\ominus}$: no. of rotations made by e- around nucleus in 1 second

$$f = \frac{1}{T} = \frac{me^4}{4h^3\varepsilon_0^2}\left(\frac{Z^2}{n^3}\right)$$

$$f = 6.59\times10^{15}\left(\frac{Z^2}{n^3}\right)\text{ Hz}$$

$$\star\ f \propto Z^2,\quad f \propto \frac{1}{n^3}$$

⑥ Effective current due to motion of $e^{\ominus}$:



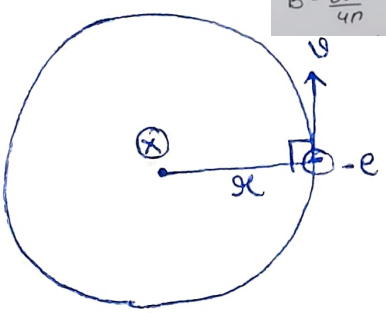
$$i = ef = \frac{me^5}{4h^3\varepsilon_0^2}\cdot\frac{Z^2}{n^3}$$

$$i = 1.05\times10^{-3}\left(\frac{Z^2}{n^3}\right)\text{ Amp}$$

$$\star\ i \propto Z^2,\quad i \propto \frac{1}{n^3}$$

⑦ Magnetic field / Magnetic induction at nucleus due to motion of $e^{\ominus}$:

$B = \frac{\mu_0}{4\pi}\frac{ev\sin 90^\circ}{r^2}$

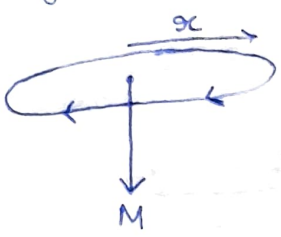


$$B = \frac{\mu_0 i}{2r} = \frac{\mu_0}{2}\,\frac{me^5}{4h^3\varepsilon_0^2}\,\frac{\pi m e^2}{h^2\varepsilon_0}\left(\frac{Z^3}{n^5}\right)$$

$$B = \frac{\mu_0 m^2 \pi e^7}{8h^5\varepsilon_0^3}\left(\frac{Z^2}{n^5}\right)$$

$$\star\ B \propto Z^2,\quad B \propto \frac{1}{n^5}$$

(8) Magnetic dipole moment (M) due to revolution of $e^{\ominus}$s :



$$M = iA = i\pi r^2 = \frac{me^3 \pi h^4 \varepsilon_0^2}{4h^3 \varepsilon_0^2 \pi^2 m^2 e^4} \cdot \frac{Z^2}{n^3} \cdot \frac{n^4}{Z^2}$$

$$\boxed{M = \left(\frac{eh}{4\pi m}\right) n}$$

independent of Z

$\frac{eh}{4\pi m}$ → Bohr Magneton

$\star$ $\boxed{M \propto n}$

(9) Energy :

i) Kinetic Energy $= \frac{1}{2}mv^2 = \frac{1}{2} m \frac{e^4}{4h^2\varepsilon_0^2} \frac{Z^2}{n^2} = \frac{me^4}{8h^2\varepsilon_0^2}\left(\frac{Z^2}{n^2}\right)$

$$\boxed{KE = 13.6\,eV\left(\frac{Z^2}{n^2}\right)}$$

ii) Potential Energy $= U = \frac{-Ze^2}{4\pi\varepsilon_0 r} = \frac{Ze^2 \pi me^2}{4\pi\varepsilon_0 h^2 \varepsilon_0}\left(\frac{Z}{n^2}\right) = -\left(\frac{me^4}{4h^2\varepsilon_0^2}\right)\frac{Z^2}{n^2}$

$$\boxed{U = -27.2\,ev\left(\frac{Z^2}{n^2}\right)} \qquad \boxed{U = -2KE}$$

iii) Total energy $= KE + PE = -\frac{me^4}{8h^2\varepsilon_0^2} \times \frac{Z^2}{n^2}$

$$\boxed{T.E = -13.6\,ev\left(\frac{Z^2}{n^2}\right) = -KE = \frac{U}{2}}$$

For H-atom :

$n=1 \Rightarrow E_1 = -13.6\,ev \times \frac{1^2}{1^2} = -13.6\,eV$

$n=2 \Rightarrow E_2 = -13.6\,ev \times \frac{1^2}{2^2} = -\frac{13.6}{4} = -3.4\,eV$

$n=3 \Rightarrow E_3 = -13.6\,ev \times \frac{1^2}{3^2} = -\frac{13.6}{9} = -1.51\,eV$

$n=4 \Rightarrow E_4 = -13.6\,ev \times \frac{1^2}{4^2} = -\frac{13.6}{16} = -0.85\,eV$

$n=5 \Rightarrow E_5 = -13.6\,ev \times \frac{1^2}{5^2} = -\frac{13.6}{25} = -0.54\,eV$

$n=6 \Rightarrow E_6 = -13.6\,ev \times \frac{1^2}{6^2} = -\frac{13.6}{36} = -0.37\,eV$

$n \to \infty \Rightarrow E \to 0$

for other single e- species, multiply the energy values with Z square

Rydberg Constant:

$$E = -\frac{me^4}{8h^2\varepsilon_0^2}\left(\frac{Z^2}{n^2}\right)$$

Multiply and divide by hc

$$E = -\frac{me^4}{8h^2\varepsilon_0^2}\left(\frac{Z^2}{n^2}\right) \times \frac{hc}{hc}$$

$$E = -\frac{me^4}{8h^3\varepsilon_0^2 c}\, hc\left(\frac{Z^2}{n^2}\right)$$

$$\boxed{E = -RhC\left(\frac{Z^2}{n^2}\right)}$$

Rydbergs Constant $= R = \frac{me^4}{8h^3\varepsilon_0^2 c}$

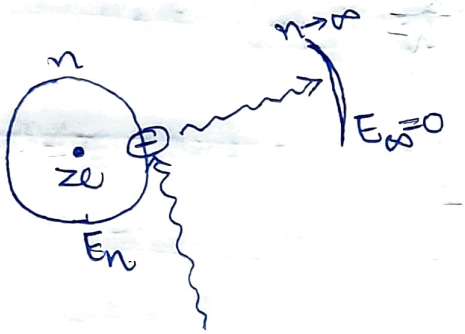
$$\boxed{R = 10967800\,m^{-1}}$$ given in problem

$$\boxed{R = \frac{13.6\,ev}{hc}}$$

$$\boxed{\frac{1}{R} = 911\times10^{-10}\,m = 911\,\text{Å}}$$

$$\therefore \boxed{E = -13.6\,ev\left(\frac{Z^2}{n^2}\right) = -RhC\,\frac{Z^2}{n^2}}$$

Ionization energy: It is the minimum energy required to transfer an $e^{\ominus}$ revolving around nucleus to $n \to \infty$.
Energy can be given by a photon incident or atomic collisions.



$$\Delta E = E_\infty - E_n = 0 - \left\{-13.6\,ev\,\frac{Z^2}{n^2}\right\}$$

$$\Delta E = 13.6\,ev\,\frac{Z^2}{n^2} = -T.E. = I.E$$

It is also the binding energy.

$$\boxed{I.E = 13.6\,ev\,\frac{Z^2}{n^2}}$$

$$\boxed{\text{Ionisation Potential} = I.P. = \frac{I.E.}{e}}$$

Q Find ionization energy and ionization potential of He^+ atom in which $e^{\ominus}$ is revolving in orbit $n=3$.

$$E_3 = -13.6\,ev \times \frac{2^2}{3^2}$$

$$= -13.6\,ev \times \frac{4}{9}$$

$$E_3 = -6.04\,ev$$

$$E_\infty = 0$$

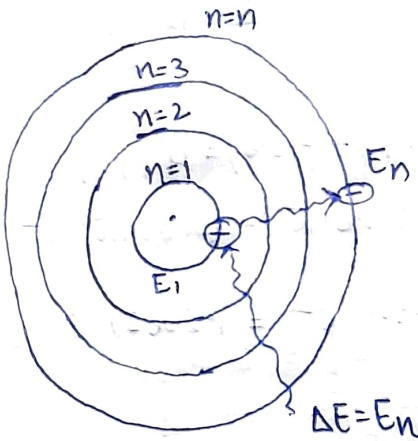
$$\Delta E = 0 - (-6.04)\,ev$$

$$\boxed{IE = 6.04\,ev}$$

$$I.P = \frac{6.04\,ev}{e}$$

$$\boxed{I.P = 6.04\,V}$$

Excitation Energy: It is defined as amount of energy required to shift $e^{\ominus}$ from ground state to the given orbit.



$$\Delta E_{ext} = E_n - E_1$$

$$= -13.6\,ev\left(\frac{Z^2}{n^2}\right) - \left\{-13.6\,ev\,\frac{Z^2}{1^2}\right\}$$

$$\boxed{E_{ext} = 13.6\,ev\,Z^2\left\{1 - \frac{1}{n^2}\right\}}$$

$$\boxed{E_{exc} = RhC\,Z^2\left\{1 - \frac{1}{n^2}\right\}}$$

$$\boxed{\text{Excitation Potential} = \frac{\text{Excitation energy}}{e}}$$

Q Find excitation energy and excitation potential of He^+ atom for stationary orbit $n = 2, 3, 4$ and 5.

$$E_1 = -13.6\,ev \times \frac{2^2}{1^2} = -54.4\,ev$$

$$E_2 = -13.6\,ev \times \frac{2^2}{2^2} = -13.6\,ev$$

$$E_3 = -13.6\,ev \times \frac{2^2}{3^2} = -6.04\,ev$$

$$E_4 = -13.6\,ev \times \frac{2^2}{4^2} = -3.4\,ev$$

$$E_5 = -13.6\,ev \times \frac{2^2}{5^2} = -2.18\,ev$$

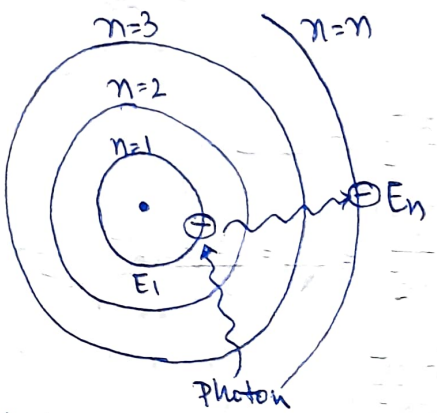
$$\Delta E_2 = E_2 - E_1 = -13.6 - (-54.4) = 40.8\,ev$$

$$\Delta E_3 = E_3 - E_1 = 48.36\,eV$$

$$\Delta E_4 = E_4 - E_1 = 51\,eV$$

$$\Delta E_5 = E_5 - E_1 = 52.22\,eV$$

Excitation of atom:



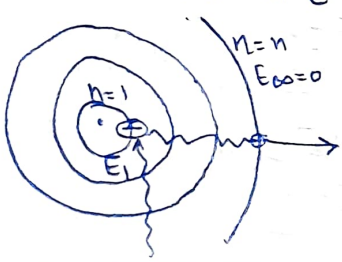
Photon having energy $\Delta E = E_n - E_1$ will be absorbed and $e^{\ominus}$ will jump from $n=1$ to $n=n$.

$$\Delta E = E_n - E_1 = \frac{hC}{\lambda}$$

$$\boxed{\Delta E(ev) = \frac{12420\,ev\text{-}\mathring{A}}{\lambda}}$$

$$\boxed{\lambda = \frac{12420}{\Delta E}\,\mathring{A}}$$

What if photons with E more than binding energy (B.E.) is incident?



$$\Delta E = \frac{hC}{\lambda} > B.E.$$

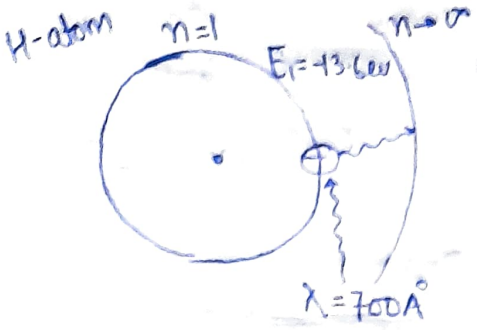
$$KE = \Delta E - (B.E.)$$

$$= \Delta E - (-E_1)$$

$$= \Delta E + E_1$$

$$\star\ \boxed{KE = \frac{hC}{\lambda} + E_1}$$

Q Find K.E. and speed of $e^{\ominus}$ when $e^{\ominus}$ comes out of atom.



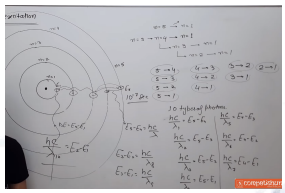
Energy of photon $= \Delta E = \frac{hc}{\lambda} = \frac{12420\ ev\text{-}Å}{700\ Å}$

$\Delta E = 17.74\ ev$

$KE = \Delta E - (-E_1) = 17.74 + (13.6) = 4.14\ eV$

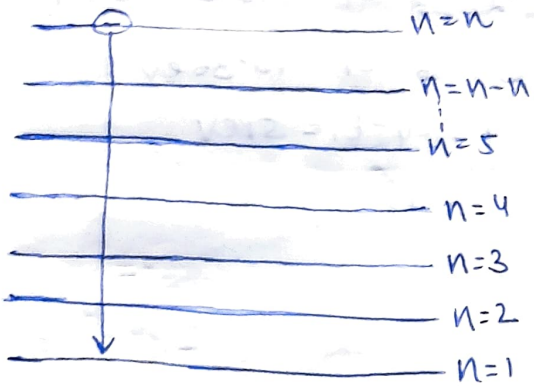
$KE = \frac{1}{2}mv^2 \Rightarrow 4.14 \times 1.6 \times 10^{-19} = \frac{1}{2} \times 9.1 \times 10^{-31} v^2$

$v = \sqrt{\frac{2 \times 4.14 \times 1.6}{9.1} \times 10^{12}} = 1.2065 \times 10^6\ m/s$



De-excitation of atom:

→ Lower Quantum number are more stable.
→ Ground state is most stable
→ If $e^{\ominus}$ is at any excited state, it will come back to ground state.
→ Energy lost will be in form of photon. $\Delta E = \frac{hc}{\lambda} = E_{n_2} - E_{n_1}$

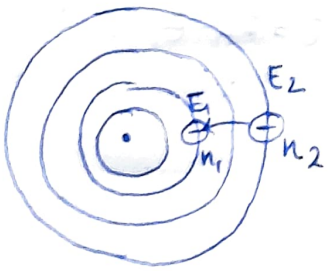


Types of photons =

$(n-1) + (n-2) + \dots + 5 + 4 + 3 + 2 + 1$

$\Sigma(n-1) = \frac{(n-1)n}{2}$

$\therefore$ Types of photons $= \frac{n(n-1)}{2} = {}^{n}C_2$



$E_2 - E_1 = \frac{hc}{\lambda}$

$= -13.6\frac{Z^2}{n_2^2} - \left\{-13.6\frac{Z^2}{n_1^2}\right\} = \frac{hc}{\lambda}$

$\frac{hc}{\lambda} = 13.6\ ev\ Z^2 \left\{\frac{1}{n_1^2} - \frac{1}{n_2^2}\right\} \Rightarrow \frac{1}{\lambda} = \frac{13.6\ ev\ Z^2}{hc}\left\{\frac{1}{n_1^2} - \frac{1}{n_2^2}\right\}$

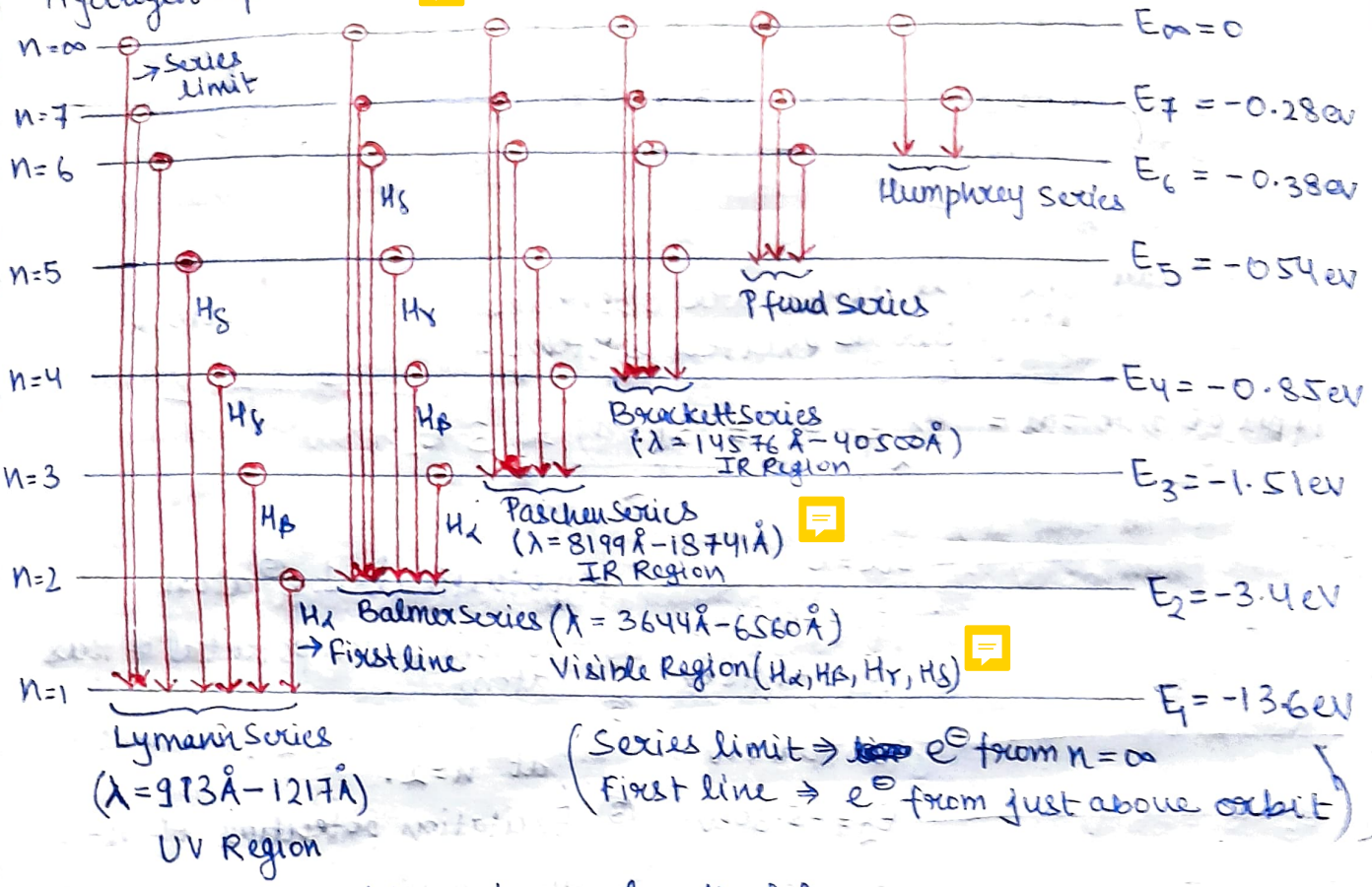
* $\boxed{\frac{1}{\lambda} = RZ^2\left\{\frac{1}{n_1^2} - \frac{1}{n_2^2}\right\}} \Rightarrow \frac{1}{\lambda} = RZ^2\left\{\frac{n_2^2 - n_1^2}{n_1^2 n_2^2}\right\}$

$\boxed{f = \frac{c}{\lambda} = RcZ^2\left\{\frac{1}{n_1^2} - \frac{1}{n_2^2}\right\}}$

$\lambda = \frac{1}{RZ^2}\left\{\frac{n_1^2 n_2^2}{n_2^2 - n_1^2}\right\}$

$\boxed{\lambda = \frac{911\ Å}{Z^2}\left\{\frac{n_1^2 n_2^2}{n_2^2 - n_1^2}\right\}}$

Hydrogen Spectrum :



(Series limit ⇒ e⊖ from n = ∞
First line ⇒ e⊖ from just above orbit)

Q Find shortest and longest wavelength of lyman series

Shortest: $E_\infty - E_1 = \frac{hc}{\lambda_{short}}$

or

$\lambda_{short} = \frac{911\text{Å}}{Z^2} \frac{n_1^2 n_2^2}{n_2^2 - n_1^2}$

$\lambda = 911\text{Å} \frac{(1^2) n_2^2}{n_2^2} = 911\text{Å}$

Longest: $n = 2$ to $n = 1$

$\lambda = 911\text{Å} \frac{(2)^2 (1)^2}{4 - 1} =$

$\lambda = 1214\text{Å}$

Q Find λ of H_γ of lyman series.

H_γ = n = 4 to n = 1 ⇒ $E_4 - E_1 = \frac{hc}{\lambda}$ ⇒ $-0.85 + 13.6 = \frac{12420}{\lambda}$

$\lambda = 974\text{Å}$

Q Find shortest and longest wavelength of balmer series.

Shortest: $E_\infty - E_1 = \frac{hc}{\lambda}$

or

$n_2 = \infty, n_1 = 2$

$\lambda = 911\text{Å} \times \frac{(2)^2 n_2^2}{n_2^2} \quad (n_2^2 >> n_1^2)$

$\lambda = 3644\text{Å}$

Longest:

$n_2 = 3$

$n_1 = 2$

$\lambda = 911\text{Å} \times \frac{(2)^2 (3)^2}{(3)^2 - (2)^2}$

$\lambda = 6560\text{Å}$

Q① Find λ of radiation required to excite the $e^{\ominus}$ in Li^{+2} from $n=1$ to $n=3$ bohr orbit.

$n_1 = 1$, $n_2 = 3$

$$\lambda = \frac{911\text{Å}}{Z^2}\left(\frac{n_1^2 n_2^2}{n_2^2 - n_1^2}\right) = \frac{911\text{Å}}{(3)^2}\left(\frac{3^2 \times 1^2}{3^2 - 1^2}\right) = \frac{911}{8}\text{ Å}$$

$$\lambda = 113.9\text{ Å}$$

② How many spectral lines are observed in the emission spectrum of the above excited system?

Types of photons $= {}^{n}C_2 = {}^{3}C_2 = \frac{3(3-1)}{2} = 3$

Q

$n=\infty$	$E_\infty = 0$
n_5	$E_5 = -0.80\text{ev}$
n_4	$E_4 = -1.45\text{ev}$
n_3	$E_3 = -3.08\text{ev}$
n_2	$E_2 = -5.30\text{ev}$
n_1	$E_1 = -15.6\text{ev}$

Hypothetical $1e^{\ominus}$ atom

① Find ionization potential of this atom

② λ short of series terminating at $n=2$.

③ Excitation potential of $n=3$

④ Wave no. of photon emitted for the transition $n=3$ to $n=1$.

① $I.E. = E_\infty - E_1 = 0 - (-15.6)\text{ev} = 15.6\text{ev}$

$$I.P. = \frac{I.E.}{e} = 15.6\text{V}$$

② $E_\infty - E_2 = \frac{hc}{\lambda}$

$$= 5.30\text{ ev} = \frac{12420\text{ eV Å}}{\lambda}$$

$$= \lambda = 2343\text{ Å}$$

③ $E_{exc} = E_3 - E_1 = -3.08 + 15.6\text{ ev}$

$E_{exc} = 12.52\text{ ev}$

$$E.P = \frac{E_{exc}}{e} = 12.52\text{ V}$$

④ Wave number $= \frac{1}{\lambda}$

Wave no. $= \frac{1}{\lambda}$

Angular wave no. $= \frac{2\pi}{\lambda}$

$$E_3 - E_1 = 12.52\text{ev} = \frac{hc}{\lambda}$$

$$= \frac{1}{\lambda} = \frac{12.52\text{ev}}{hc} = \frac{12.52\text{eV}}{12420\text{ eV-Å}} = \frac{12.52\text{ ev} \times 10^{10}}{12420\text{ ev} \times m}$$

$$\frac{1}{\lambda} = 1.00805 \times 10^7\ m^{-1}$$

Q A H-atom is excited through EM radiation such that it emits visible light during de-excitation. Find the maximum of radiation by which it is possible.

During de-excitation, visible light is emitted ⇒ Balmer series ⇒ $e^{\ominus}$ comes to $n=2$.
For max^m wavelength, we need least energy to excite $e^{\ominus}$ and make it reach $n=3$.

$n_1=1$
$n_2=3$

$$\lambda = \frac{911\text{Å}}{Z^2}\frac{n_1^2 n_2^2}{n_2^2-n_1^2} = 911\text{Å} \times \frac{9}{8} = 1024\text{Å}$$

Q Hydrogen gas is excited by radiation of $\lambda = 974$Å. How many spectral lines are possible in resulting emission spectrum. Also find longest wavelength.

Let $e^{\ominus}$ reach n^{th} orbit on excitation.

$$E_n - E_1 = \frac{hc}{\lambda}$$

$$= \frac{-13.6\text{ eV}}{n^2} + 13.6\text{eV} = \frac{12420\text{ eVÅ}}{974\text{Å}}$$

$$= 13.6\text{ eV} - \frac{12420}{974} = \frac{13.6}{n^2}$$

$$= \frac{1}{n^2} = 1 - \frac{12.75}{13.6} = 0.062$$

$$= n^2 = \frac{1}{0.62} = 16.02 \Rightarrow \boxed{n=4}$$

∴ No. of spectral lines possible:
$${}^4C_2 = \frac{4\times3}{2} = 6 \text{ lines.}$$

② Longest λ will be when $e^{\ominus}$ comes from $n=4$ to $n=3$.

$$\lambda = \frac{911\text{Å}}{1^2} \times \frac{4^2\times3^2}{4^2-3^2}$$

$$\lambda = 18820\text{Å}$$

3.) the smallest wavelength emitted:
A. it will same as the excitation wavelength

Q UV light of $\lambda = 690$Å is incident on a H-atom in ground state. Find KE and speed of $e^{\ominus}$ emitted.

$$E = \frac{hc}{\lambda} = \frac{12420\text{ eVÅ}}{690\text{Å}} = 18\text{eV}$$

$$KE = E - (B.E.)$$
$$= 18\text{eV} - 13.6\text{eV}$$
$$= 4.4\text{ eV}$$

Now, $KE = \frac{1}{2}mv^2$

$$4.4\times1.6\times10^{-19} = \frac{1}{2}\times9.1\times10^{-31}\times v^2$$

$$v = \sqrt{\frac{2\times4.4\times1.6\times10^{-19}}{9.1\times10^{-31}}}$$

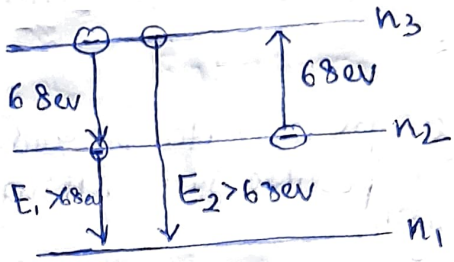
$$v = 1.24\times10^6\text{ m/s}$$

** A gas of H-like atom can absorb radiation of 68 eV. Consequently the atom emit radiations of only 3 different λ. All the λ are equal or smaller than that of absorbed photon. Find ① Initial state of atom ② Find the atom ③ Find the min^m λ of emitted radiations. ④ Find ionisation energy of atom.

① Let the $e^\ominus$ excite upto n^{th} orbit

$${}^nC_2 = 3 \Rightarrow \frac{n(n-1)}{2} = 3$$

$\boxed{n=3}$



Now given that emitted λ are equal or smaller than 68 eV. If $e^\ominus$ was initially at ground state then while returning it could come to n_2 from n_3 in which case λ would have been greater than 68 eV. So $e^\ominus$ is not at ground state.

∴ This scenario is only possible when initial state is $n=2$

② $E_3 - E_2 = 68$ eV

$$= \frac{-13.6}{9}Z^2 - \left\{\frac{-13.6\,Z^2}{4}\right\} = 13.6\,Z^2\left\{\frac{1}{4} - \frac{1}{9}\right\} = 68$$

$$= 13.6 \times \frac{5}{36} \times Z^2 = 68 \Rightarrow Z^2 = \frac{68 \times 36}{5 \times 13.6} = 36$$

∴ $Z=6 \Rightarrow$ Carbon atom

③ Min^m λ is when $e^\ominus$ comes from n_3 to n_1.

$$\lambda = \frac{911\text{Å}}{36} \times \frac{1^2 \times 3^2}{3^2 - 1} = \frac{911}{436} \times \frac{9}{8} = \frac{911}{32} = 28.46\text{Å}$$

④ $$I.E = +13.6\frac{Z^2}{n^2}$$

Here $n=1$

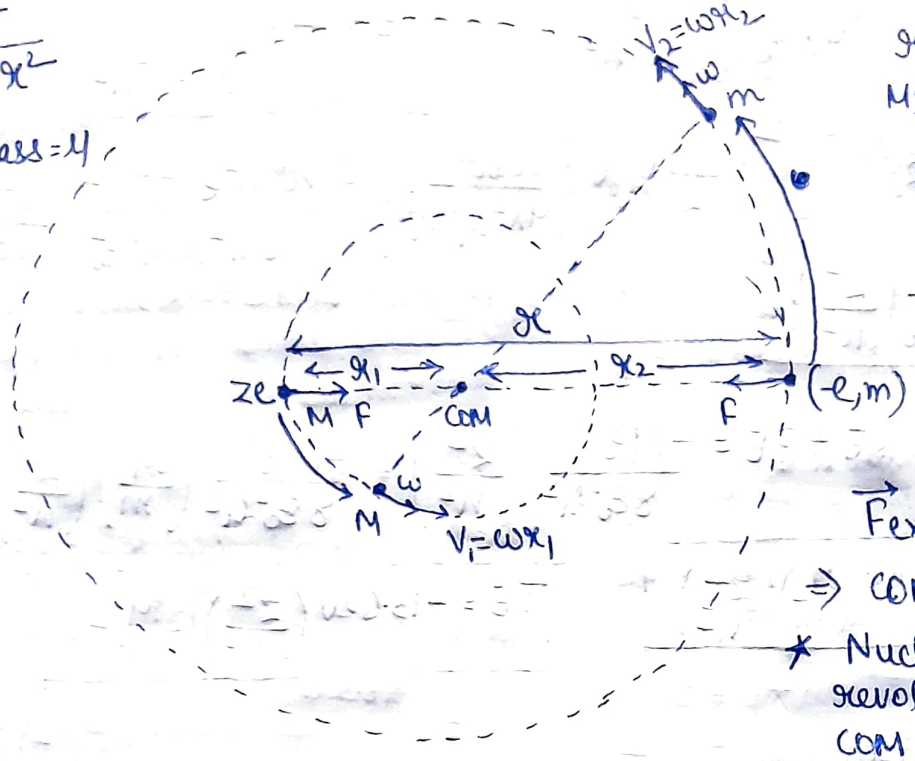
$$IE = +13.6 \times 36 = 489.6 \text{ eV}$$

Effect of motion of nucleus: M = Mass of nucleus
r = sep. b/w nucleus and $e^{\ominus}$.

$$F = \frac{ze^2}{4\pi\varepsilon_0 r^2}$$

Reduced mass = μ

$$\mu = \frac{mM}{m+M}$$



$r_1 + r_2 = r$ — (1)

$Mr_1 = mr_2$ — (2)

$$r_1 = \frac{mr}{M+m}$$

$$r_2 = \frac{Mr}{M+m}$$

$\vec{F}_{ext} = 0$

$\Rightarrow$ COM - Fixed

* Nucleus and $e^{\ominus}$ will revolve around the COM with equal angular velocity (ω).

Radius:

NL $\Rightarrow F = ma$

$$\frac{ze^2}{4\pi\varepsilon_0 r^2} = m\omega^2 r_2$$

$$= \frac{ze^2}{4\pi\varepsilon_0 r^2} = \frac{m\omega^2 Mr}{M+m} = \mu\omega^2 r \quad \text{— (1)}$$

AMQR, $L = \frac{nh}{2\pi}$

$$L = I\omega = \frac{nh}{2\pi}$$

$$\mu r^2 \omega = \frac{nh}{2\pi} \quad \text{— (2)}$$

$$I = Mr_1^2 + mr_2^2 = \mu r^2$$

Put ω in (1) from (2)

$$\frac{ze^2}{4\pi\varepsilon_0 r^2} = \mu r \left\{ \frac{n^2h^2}{4\pi^2\mu^2 r^2 r} \right\} \Rightarrow \frac{ze^2}{\varepsilon_0} = \frac{n^2h^2}{\pi\mu r}$$

$$r = \frac{n^2h^2\varepsilon_0}{ze^2\pi\mu} \Rightarrow r = \frac{h^2\varepsilon_0}{e^2\pi\mu}\left(\frac{n^2}{z}\right)$$

Multiply and divide by m

$$r = \frac{h^2\varepsilon_0}{e^2\pi m}\left(\frac{m}{\mu}\right)\left(\frac{n^2}{z}\right)$$

$$\therefore \boxed{r = 0.529\,\text{Å}\left(\frac{n^2}{z}\right)\left(\frac{m}{\mu}\right)}$$

$$\boxed{r = 0.529\,\text{Å}\left(\frac{M+m}{M}\right)\left(\frac{n^2}{z}\right)}$$

Energy:

$$PE = \frac{-ze^2}{4\pi\varepsilon_0 r} = \frac{-e^2 e^2 \pi \mu}{4\pi\varepsilon_0 h^2 \varepsilon_0}\left(\frac{z^2}{n^2}\right)$$

$$\boxed{PE = \frac{-\mu e^4}{4\varepsilon_0^2 h^2}\left(\frac{z^2}{n^2}\right)}$$

$$KE = \frac{1}{2} I\omega^2 = \frac{1}{2}\mu r^2 \omega^2 = \frac{1}{2}\mu r^2 \left\{\frac{n^2 h^2}{4\pi^2 \mu^2 r^4}\right\} = \frac{n^2 h^2 e^4 \pi^2 \mu^2}{8\pi^2 \mu h^2 \varepsilon_0^2}\left(\frac{z^2}{n^2}\right)$$

$$\boxed{KE = \frac{\mu e^4}{8h^2\varepsilon_0}\left(\frac{z^2}{n^2}\right)}$$

$$\text{Total energy} = KE + PE = \frac{-\mu e^4}{8\varepsilon_0^2 h^2}\left(\frac{z^2}{n^2}\right) - \frac{-me^4}{8\varepsilon_0^2 h^2}\left(\frac{\mu}{m}\right)\left(\frac{z^2}{n^2}\right)$$

$$\boxed{T.E = -13.6\,ev\left(\frac{\mu}{m}\right)\left(\frac{z^2}{n^2}\right)} \qquad \boxed{TE = -13.6\,ev\left(\frac{z^2}{n^2}\right)\left(\frac{m_1}{M+m}\right)}$$

Q Positronium atom

↳ Positron + $e^{\ominus}$

→ antiparticle of $e^{\ominus}$

→ $m = m_e = 9.1\times10^{-31}$ kg

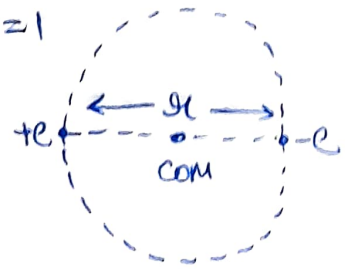
→ charge $= +e = 1.6\times10^{-19}$ C

① Find radii of possible orbits.

② Find energy of different quantum levels.

$ze = e$

$z = 1$



① $\mu = \frac{m\cdot m}{m+m} = \frac{m}{2}$

$r = 0.529\,\text{Å}\left(\frac{n^2}{z}\right)\left(\frac{m}{\mu}\right)$

$r = 0.529\,\text{Å}\left(\frac{n^2}{1}\right)\left(\frac{m}{m/2}\right)$

$\boxed{r = 1.058\,\text{Å} \times n^2}$

② $E = -13.6\left(\frac{z^2}{n^2}\right)\left(\frac{\mu}{m}\right) ev$

$E = -13.6ev\left(\frac{1}{n^2}\right)\left(\frac{m/2}{m}\right) ev$

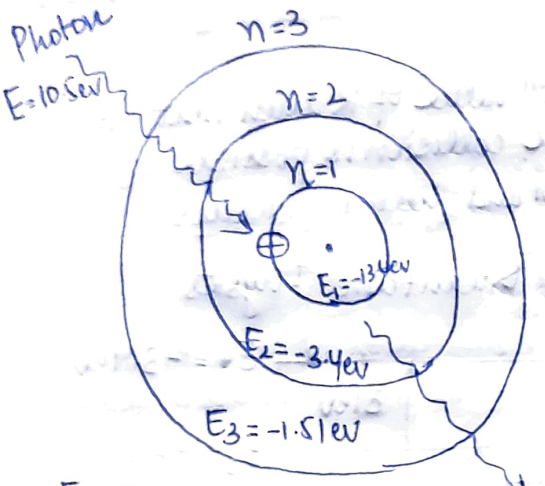
$E = -13.6\,ev\left(\frac{1}{n^2}\right)\left(\frac{1}{2}\right) ev$

$\boxed{E = -\frac{6.3}{n^2}\,ev}$

Atomic Collision :

(I) Collision of atom with an electron :

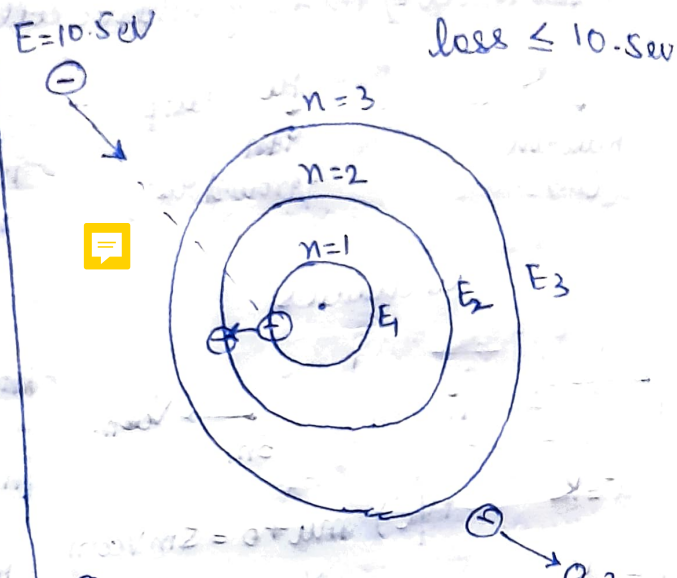
→ We can ignore momentum of fastly moving $e^{\ominus}$ when it collides with an atom but we have to consider its K.E.



$E_2 - E_1 = 10.2ev$

$E_3 - E_1 = 12.09ev$

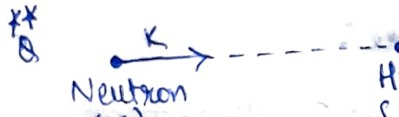
As energy of photon is not E_2-E_1 or E_3-E_1, it will not be absorbed and the photon will come out with same E



$e^{\ominus}$ will strike the other $e^{\ominus}$ and will give it energy 10.2ev to move to n=2 and with the left energy it will come out.

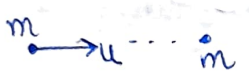
(II) Collision between atoms or Collision b/w atom and neutron :

→ In atomic collisions, any energy loss is used to excite the atom.



Find type of collision if K = 10ev

Mechanics



$\frac{1}{2}mu^2 = K$

using M/C

$\Rightarrow mu + 0 = 2m\,V_{com}$

$V_{com} = v/2$

$K_i = K$

$K_f = \frac{1}{2}2m\left(\frac{v}{2}\right)^2 = \frac{mv^2}{4} = K/2$

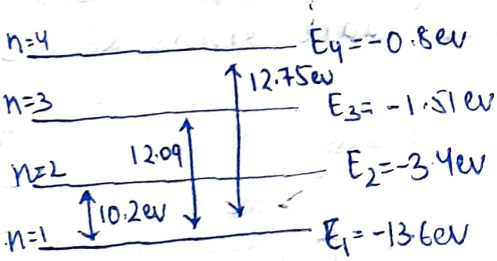
Energy loss = $[0, K/2]$ → If $K=0 \Rightarrow$ elastic

If $K = \frac{K}{2} \Rightarrow$ Perfectly inelastic

If $K \in (0, \frac{K}{2}) \Rightarrow$ inelastic

Max^m loss = $\frac{K}{2}$

Quantum Physics



Energy loss = {0, 10.2 ev, 12.09 ev, 12.75 ev ...}

Mechanics ⇒ loss = [0, 5 ev]

Common value is 0.

∴ Energy loss is 0 ⇒ elastic collision

(ii) If K = 20.4 ev

Mech ⇒ loss = [0, 10.2 ev]

Quantum ⇒ loss = {0, 10.2 ev, 12.09 ev, ...}

Common value = {0, 10.2 ev}

If loss = 0 ⇒ elastic collision

If loss = 10.2 ev ⇒ Perfectly inelastic collision

(iii) If K = 21 ev

Mech ⇒ loss = [0, 10.5 ev]

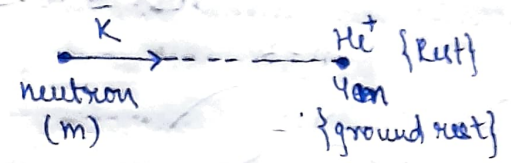
Quantum ⇒ loss = {0, 10.2 ev, 12.09 ev, ...}

Common value ⇒ {0, 10.2 ev}

If loss = 0 ⇒ elastic collision

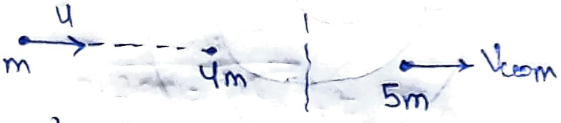
If loss = 10.2 ev ⇒ inelastic collision

Q



Find min^m value of K such that perfectly inelastic collision is possible b/w neutron and helium nucleus.

Mechanics



$\frac{1}{2}mu^2 = K$

$K_i = K$

$K_f = \frac{1}{2}(5m)\left(\frac{u}{5}\right)^2 = \frac{mu^2}{10} = \frac{K}{5}$

M/C ⇒ $mu + 0 = 5mV_{com}$

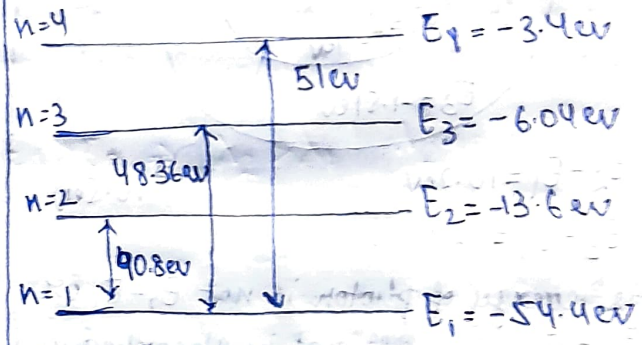
$V_{com} = u/5$

Max^m loss = $K - \frac{K}{5} = \frac{4K}{5}$

loss = $[0, \frac{4K}{5}]$

For perfectly inelastic collision ⇒ loss = $\frac{4K}{5}$

Quantum Physics



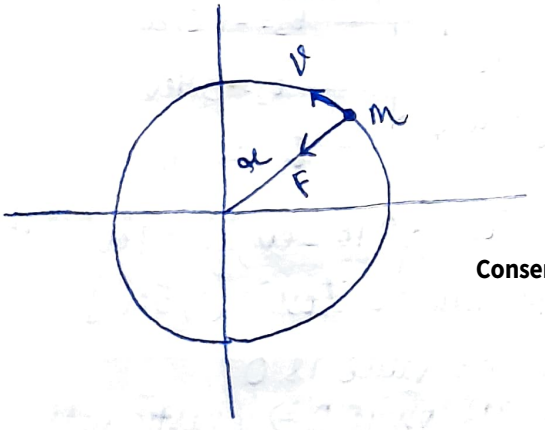
loss = {0, 40.8, 48.36, 51 ...}

For min^m value of K ⇒

$\frac{4K}{5} = 40.8$ ⇒ **K = 51 ev**

Hypothetical Model's based on Bohr's Quantization rule:

Q Potential energy in a region is given as $U = kr^2$, r is distance from origin. Particle of mass m is in this region performing circular motion. If particle follows angular momentum quantisation rule, then find its possible radii, speeds, KE and total energy.



As potential at each point on circle is same (as r same) so it is an equipotential surface so force F must be ⊥ to surface ie radially inwards or outwards

Now as r↑, U↑

Conservative Force will be from higher U to lower U.

∴ Force is radially inwards.

$F = -\frac{dU}{dr} = -2kr$ { -ve represents inwards }

NL: $F = ma$

$2kx = \frac{mv^2}{x}$ — ①

AMQR: $L = \frac{nh}{2\pi}$

$mvx = \frac{nh}{2\pi}$ — ②

From ① and ②

$2kx^2 = m\left\{\frac{n^2h^2}{4\pi^2m^2x^2}\right\}$

$\Rightarrow x^4 = \frac{h^2}{8\pi^2mk}\, n^2$

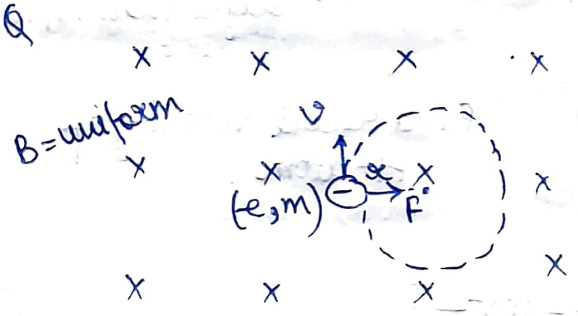
$\therefore \boxed{x = \left(\frac{h}{8\pi^2mk}\right)^{1/4}\sqrt{n}}$

Now, $v = \frac{nh}{2\pi mx} = \frac{nh}{2\pi m}\left(\frac{8\pi^2mk}{h^2}\right)^{1/4}\frac{1}{\sqrt{n}} = \frac{h}{2\pi m}\left(\frac{8\pi^2mk}{h^2}\right)^{1/4}\sqrt{n}$

$KE = \frac{1}{2}mv^2 = \frac{1}{2}m\left\{\frac{h^2n}{4\pi^2m^2}\sqrt{\frac{8\pi^2mk}{h^2}}\right\}$ $\quad PE = kx^2 = kn\left(\frac{h^2}{8\pi^2mk}\right)^{1/2}$

$\boxed{TE = KE + PE}$

Q



This $e^\ominus$ will perform UCM only in the paths which follow Bohr's postulate of AMQR.
Then find ① Possible radii of path
② Speeds of $e^\ominus$ ⑤ T.E.
③ KE of $e^\ominus$
④ PE of $e^\ominus$

NL: $F = ma$

$evB\sin 90 = \frac{mv^2}{x}$

$x = \frac{mv}{eB}$ — ①

AMQR: $L = \frac{nh}{2\pi}$

$mvx = \frac{nh}{2\pi}$ — ②

From ① and ②

$eBx^2 = \frac{nh}{2\pi}$

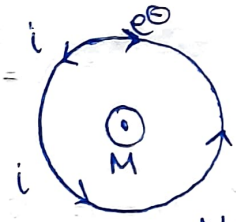
$\boxed{x = \sqrt{\frac{nh}{2\pi eB}}}$

$\boxed{v = \frac{1}{m}\sqrt{\frac{nheB}{2\pi}}}$

$KE = \frac{1}{2}mv^2 = \frac{1}{2}m\,\frac{nheB}{2\pi}\times\frac{1}{m^2}$

$\boxed{KE = \frac{nheB}{4\pi m}}$

④



$i = \frac{e}{T} = \frac{ev}{2\pi x}$

$M = iA = \frac{ev}{2\pi x}\times\pi x^2 = \frac{evx}{2}$

$U = -MB\cos\theta = -\left\{\frac{evx}{2}\right\}B\cos\pi$

$U = \frac{evxB}{2} = \frac{eB}{2}\left\{\frac{nh}{2\pi m}\right\}$

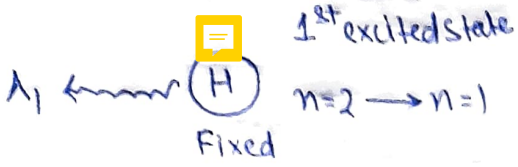
$\boxed{U = \frac{nheB}{4\pi m}}$

$\boxed{TE = \frac{nheB}{2\pi m}}$

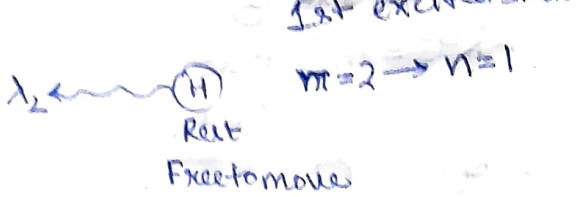
Recoil speed of H-atom on emission of photon:

Q

Case I



Case II



Which is larger wavelength?

Case I:

$E_2 = -3.4 eV$

$E_1 = -13.6 eV$

$E_2 - E_1 = \frac{hc}{\lambda_1}$

$$\boxed{10.2 eV = \frac{hc}{\lambda_1}} = E_A$$

Case II.

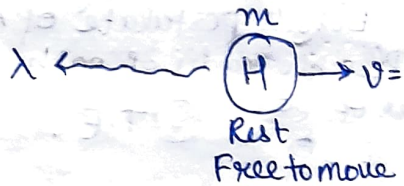


$E_B = 10.2 eV = \frac{hc}{\lambda_2} + \frac{1}{2}mv^2$

H-atom also moves with v so that final momentum of the system is also zero.

As $E_A > E_B \Rightarrow \boxed{\lambda_2 > \lambda_1}$ λ_2 is slightly larger than λ_1.

Q



1st excited state $n=2 \rightarrow n=1$

Find recoil velocity of H-atom.

$E/C \Rightarrow E_2 - E_1 = \frac{hc}{\lambda} + \frac{1}{2}mv^2 \Rightarrow 10.2 eV = \frac{hc}{\lambda} + \frac{1}{2}mv^2$ — ①

$M/C = \frac{h}{\lambda} = mv$ — ②

From ① and ②

$= 10.2 eV = mvc + \frac{1}{2}mv^2$

$= \boxed{\frac{mv^2}{2} + mcv - 10.2 eV = 0}$

$$v = \frac{-mc \pm \sqrt{m^2c^2 + 4\frac{m}{2} \times 10.2 eV}}{m}$$

Reject -ve ~~as v is very small~~

$$v = -c + \sqrt{c^2 + \frac{2 \times 10.2 eV}{m}}$$

$$v = -c + c\sqrt{1 + \frac{2 \times 10.2 \times 1.6 \times 10^{-19}}{1.67 \times 10^{-27} \times 9 \times 10^{16}}}$$

$$v = -c + c\left\{1 + \frac{2 \times 10.2 \times 1.6}{1.67 \times 9} \times 10^{-8}\right\}^{1/2}$$

Using binomial approx $\Rightarrow (1+x)^n = 1 + nx$

$$v = -c + c\left\{1 + \frac{10.2 \times 1.6}{1.67 \times 9} \times 10^{-8}\right\}$$

$$= v = -c + c + 3 \times 10^8 \times \frac{10.2 \times 1.6}{1.67 \times 9} \times 10^{-8}$$

$$= v = \frac{3 \times 10.2 \times 1.6}{1.67 \times 9}$$

$$\boxed{v = 3.26 \text{ m/s}}$$

↳ Recoil spee

$E = \frac{p^2}{2m}$?

$10.2 eV = \left(\frac{hc}{\lambda}\right)$

$10.2 eV = mvc$

$v = \frac{10.2 eV}{mc}$

$= \frac{10.2 \times 1.6 \times 10^{-19}}{1.67 \times 10^{-27} \times 3 \times 10^8}$

$= \frac{10.2 \times 1.6}{1.67 \times 3}$